

charge current were changed between 0.2-2 Torr and 10-30 mA respectively. The electron temperature (T_e) was measured with a Langmuir probe. The excitation temperature (T_{exc}) was estimated from the intensity ratio of the emission lines H_γ and H_δ . It was measured that this temperatures depends strongly on the amount of H_2O due to the vibrational collisions between electrons and H_2O molecules. When the water vapor concentration was optimal, we observed a significant decreasing of T_e and T_{exc} in comparison with pure hydrogen case (4-5 times). It is essential for the H^- generation because the cross section for the reaction of dissociative attachment $H_2^+ + e = H^+ + H^-$ is inversely proportional with the electron temperature.

2FP10

Diffusion Processes and Sheath Structure in the Near-Anode Region of an Arc Discharge in the Presence of a Transverse Magnetic Field
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Understanding the nature of processes occurring at the electrodes of DC plasmotrons is an important aspect of the field of electric propulsion. Such an understanding yields information on engine lifetime which is related to erosion processes occurring at the electrodes and on engine efficiency which is related to energy loss phenomena occurring at the electrodes. This study attempts to understand the nature of power deposition into the anode of an argon arc discharge in the presence of a transverse magnetic field. The ratio of the transverse diffusion coefficient to the parallel diffusion coefficient, $D_\perp/D_\parallel$, in the near-anode region is measured experimentally via electrostatic probes. In addition, the behavior of this parameter is correlated with measured anode sheath voltage which has been shown to be directly related to energy loss phenomena occurring in certain electric propulsion devices. Along with the diffusion coefficient data, spatial profiles of electron and ion number density, electron temperature, and plasma potential profiles have also been measured. In addition to the electrostatic probe measurements, spatially resolved spectroscopic measurements have also been obtained in the near-anode region. Spatial variations in the intensity of argon ion emission lines near the surface of the anode have been obtained. Such data provide a qualitative measure of the spatial extent and behavior in general of the anode sheath as a function of magnetic field and discharge current. Data obtained from the above mentioned studies shed light on why the measured large positive anode fall voltages develop.

2FP11

Static Current-Voltage Characteristics for Radio-Frequency Induction Discharge

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The aim of this work was to obtain experimentally such characteristic of Radio-Frequency Induction Discharge (RFID) that can play the role of its current-voltage characteristic (CVC) and to explain nature of current and voltage jumps arising in RF coil at exciting of discharge [Smith H., Lynch W.A., Hilbery N., Electrodeless discharge in mercury vapor, Phys. Rev., 1931. v.37, May, p.1091-1101].

Experiments were made in quartz 5.5, 11, 20 cm diam tubes with outer RF coil at pressures 10-100 mTorr, at frequency 13.56 MHz and discharge power to 500 W. In case of outer coil as analogue of discharge voltage it's convenient to use the value of the RF voltage U_R , induced around outer perimeter of discharge tube.

Inductive discharge current is closed in plasma volume therefore it is impossible to measure its value immediately. As an analogue of discharge current we use ohmic component I_Ω of coil RF current because it is simply expressed through parameters of RF circuit and ohmic part of generator output current which can be easily measured. At pressures above 10^{-2} Torr it's true, because frequency of electron-atomic collisions exceeds RF field frequency and ohmic component predominates in electrical impedance of RFID.

The dependencies U_R from I_Ω were built as analogues of static CVC for RFID. The generator with specially increased output resistance was used in this experiments. It has allowed to obtain smoothing (without jumps) dependence U_R upon I_Ω in a whole range of these parameters values starting from exciting point to steady-state regime. Discharge voltage has maximum value at the excite moment and decreases to some definite level when discharge current I_Ω grows. This level depends on gas pressure and tube diameter. Thus, this CVC of RFID is like to stabilitrone one. Dimensions and slope of initial downward part of characteristic depend on gas pressure and decrease at its reducing.

It is evident that current and voltage jumps arising at exciting of discharge are due to low output resistance of standard generators and negative slope of initial part of CVC.

Three sets of such dependencies for different pressures were obtained for each diameter of tubes. The influence of different metal electrodes placed into discharge volume on CVC's shape has been studied also.

Experimental results can explain the behaviour of HFI discharge as a load of RF generator and give data for calculation of RF circuit.